

ENGLISH EDITION

ESP32-2432S0248R-PLUS

2.8-inch Resistive Touch Display Development Board

Powered by the ESP32-WROOM-32UE Module

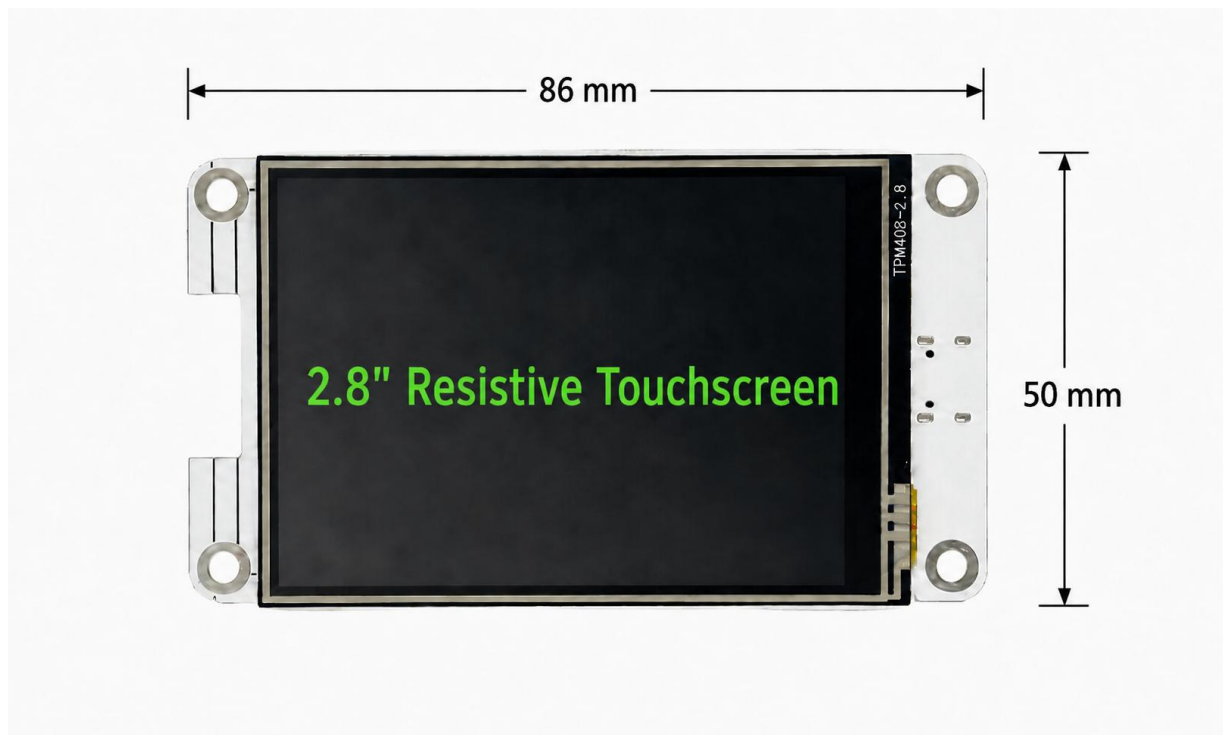
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Figure 1 Front view with dimensions

2.8-inch resistive touch 240 × 320 resolution	ESP32-WROOM-32UE 2.4 GHz Wi-Fi + Bluetooth / BLE module
ST7789 + BGR Initialize the ST7789 LCD driver in BGR color order	XPT2046 touch controller Also referred to as XP2046 in some documents

1 Product Overview

ESP32-2432S0248R-PLUS is a 2.8-inch color touch display development board based on the ESP32-WROOM-32UE wireless module. It integrates an LCD, touch input, a TF card socket, USB Type-C, UART, a battery port and several expansion interfaces. Typical uses include IoT display terminals, smart-home control panels, HMI screens, teaching labs and embedded projects.

2 Key Features

Item	Specification
Main module	ESP32-WROOM-32UE
Wireless	2.4 GHz Wi-Fi 802.11 b/g/n; Bluetooth / BLE
Display	2.8-inch TFT LCD, 240 × 320
Touch	Resistive touch, XPT2046 controller

Item	Specification
LCD driver	ST7789, BGR color-order initialization
Expansion	UART, TF card, GPIO/SPI expansion, audio port

3 Basic Specifications

Item	Specification
Model	ZLX-ESP32-1
Main module	ESP32-WROOM-32UE
Screen size	2.8 inch
Screen type	TFT LCD + resistive touch
Resolution	240 × 320
LCD driver	ST7789
Display init	ST7789 BGR color-order initialization
Touch controller	XPT2046 / XP2046
Board size	86 mm × 50 mm
Power	USB Type-C 5 V input; external 3.3 V battery port
Logic level	3.3 V
Resources	https://www.zlxchina.com/

4 ESP32-WROOM-32UE Module Parameters

Item	Specification
Module	ESP32-WROOM-32UE
CPU core	Xtensa dual-core 32-bit LX6
Clock frequency	Up to 240 MHz
ROM	448 KB
SRAM	520 KB
RTC SRAM	16 KB
Flash	Typical board configuration: 4 MB. The actual batch may vary.
Wi-Fi	2.4 GHz, 802.11 b/g/n
Bluetooth	Bluetooth BR/EDR and Bluetooth Low Energy
Antenna	On-board PCB antenna
Operating voltage	Module supply 3.0 V to 3.6 V; board logic level 3.3 V
Module size	18.0 mm × 25.5 mm × 3.1 mm

5 Display and Touch

This product uses a 2.8-inch resistive touch screen and an XPT2046 touch controller (also written XP2046 in some documents). The LCD driver is ST7789 and should be initialized with BGR color order. If red and blue appear swapped, check the TFT_RGB_ORDER / BGR setting in the display driver.

6 Recommended Initialization

These macros match the GPIO map in Section 9 and can be used directly with drivers such as TFT_eSPI.

```
// TFT display / ST7789
#define ST7789_DRIVER
#define TFT_WIDTH 240
#define TFT_HEIGHT 320
#define TFT_MISO 12
#define TFT_MOSI 13
#define TFT_SCLK 14
#define TFT_CS 15
#define TFT_DC 2
#define TFT_RST -1
#define TFT_BL 21
#define TFT_BACKLIGHT_ON HIGH
#define TFT_RGB_ORDER TFT_BGR // ST7789 BGR initialization
// Touch / XPT2046
#define TOUCH_IRQ 36
#define TOUCH_MISO 39
#define TOUCH_MOSI 32
#define TOUCH_CS 33
#define TOUCH_CLK 25
```

7 Interfaces

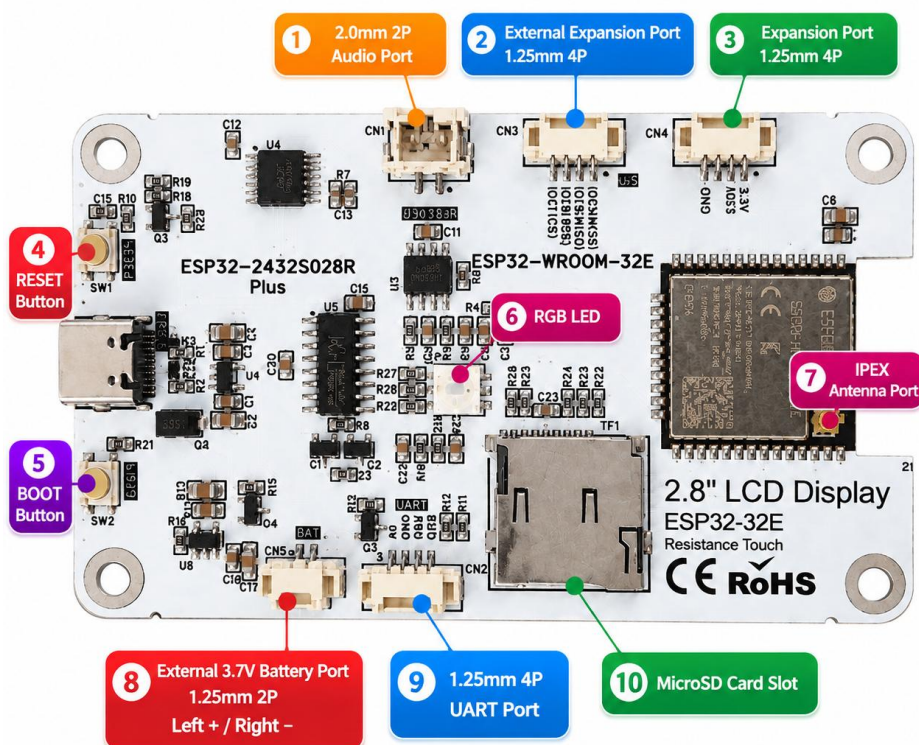


Figure 2 Back-side interface labels

Interface	Description
2.0P audio port	Reserved audio port for input/output or amplifier expansion. The exact function depends on the firmware.
External expansion port	Reserved expansion interface for SPI/GPIO peripherals, sensors or custom modules.
Expansion port	Reserved 3.3 V, GND and GPIO signals for external low-power modules.
External 3.3 V battery port	1.25P connector, left positive / right negative. Do not reverse polarity. Do not apply 5 V.
1.25P UART port	UART communication and debug port.
TF card slot	MicroSD/TF socket for data, images or configuration files.

8 Package Contents

Bare-board version:

- 1 × development board
- 1 × USB Type-C data cable
- 1 × IPEX1-to-SMA adapter
- 1 × 3 dBi rubber-duck antenna

9 GPIO Map

The following map is taken from the GPIO assignment of this product. Confirm pin direction, voltage and multiplexed functions before writing firmware. GPIO34, GPIO36 and GPIO39 are input-only and cannot be used as outputs.

Module	Signal / Silkscreen	ESP32 GPIO	Code macro	Direction	Notes
MCU	RESET	EN	LCD_RST=-1	Reset in	Tied to ESP32 EN; no GPIO used
LCD/ST7789	LCD_SCK	GPIO14	LCD_SCK=14	Output	LCD SPI clock, VSPI
LCD/ST7789	LCD_MISO	GPIO12	LCD_MISO=12	Input	LCD SPI MISO; ST7789 does not read the panel by default
LCD/ST7789	LCD_MOSI	GPIO13	LCD_MOSI=13	Output	LCD SPI MOSI
LCD/ST7789	LCD_RS / DC	GPIO2	LCD_RS=2	Output	LCD data/command select
LCD/ST7789	LCD_CS	GPIO15	LCD_CS=15	Output	LCD chip select
LCD/ST7789	LCD_BL	GPIO21	LCD_BL=21; BACKLIGHT_PIN=21	Output	LCD backlight
Touch/XPT2046	RTP_IRQ / SENSOR_VP	GPIO36	RTP_IRQ=36; TOUCH_IRQ=36	Input	Touch interrupt, input-only pin
Touch/XPT2046	RTP_DOUT / SENSOR_VN	GPIO39	RTP_DOUT=39; TOUCH_MISO=39	Input	Touch SPI DOUT/MISO, input-only pin
Touch/XPT2046	RTP_DIN	GPIO32	RTP_DIN=32; TOUCH_MOSI=32	Output	Touch SPI DIN/MOSI
Touch/XPT2046	RTP_CS	GPIO33	RTP_CS=33; TOUCH_CS=33	Output	Touch chip select
Touch/XPT2046	RTP_SCK	GPIO25	RTP_SCK=25; TOUCH_CLK=25	Output	Touch SPI clock

9 GPIO Map (continued)

Module	Signal / Silkscreen	ESP32 GPIO	Code macro	Direction	Notes
Battery	BAT_ADC	GPIO34	BAT_ADC=34; BAT_ADC_PIN=34	Input	Battery voltage ADC, input-only pin
RGB LED	RGB_R	GPIO17	RGB_LED_R=17	Output	On-board RGB red (RGB = 17/22/16)
RGB LED	RGB_G	GPIO22	RGB_LED_G=22	Output	On-board RGB green (RGB = 17/22/16)
RGB LED	RGB_B	GPIO16	RGB_LED_B=16	Output	On-board RGB blue (RGB = 17/22/16)
Audio	AUDIO_IN	GPIO26	AUDIO_IN=26	Input/ADC	Reserved audio input
Audio	AUDIO_EN	GPIO4	AUDIO_EN=4	Output	Reserved audio / amplifier enable
Exp. SPI	SPI_CS	GPIO27	SPI_CS=27	Output	Reserved expansion SPI chip select
SD card	SD_MOSI	GPIO23	SD_MOSI=23	Output	Reserved SD SPI MOSI
SD card	SD_MISO	GPIO19	SD_MISO=19	Input	Reserved SD SPI MISO
SD card	SD_SCK	GPIO18	SD_SCK=18	Output	Reserved SD SPI clock
SD card	SD_CS	GPIO5	SD_CS=5	Output	Reserved SD card chip select

10 Important Notes

Item	Note
GPIO level	Board GPIO logic is 3.3 V. Do not connect 5 V signals directly.
Input-only pins	GPIO34, GPIO36 and GPIO39 are input-only and cannot be used as outputs.

Item	Note
Battery port	The external 3.3 V battery port is left positive / right negative. Do not reverse polarity or apply 5 V.
Touch calibration	Recalibrate touch after the display is rotated or the coordinate direction changes.
Display color	If red and blue are swapped, check the ST7789 BGR/RGB initialization.
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Warning: the external battery port is 3.3 V only, left positive and right negative. Reverse polarity or 5 V input can damage the board.

11 Typical Applications

Smart-home control panel	IoT data display terminal
Industrial equipment status display	Sensor data display
Wi-Fi / Bluetooth control terminal	Educational development board

12 Technical Support

Development files, sample programs, firmware, the GPIO map and updates are available at:

Resource download <https://www.zlxchina.com/>